Supplementary Material: Appendices A, B, and C

Technical Foundations, Empirical Telemetry, and Reproducibility Specifications

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Appendix A: Technical Architecture & Mathematical Formulations

A.1 Mathematical Formulation of Decision-Uncertainty Routing

In credit card fraud detection, conventional routing policies stratify transactions by currency volume (Transaction Amount), assuming high-value transactions represent dominant risk. However, organized fraud rings routinely exploit low-value authorization testing ($< \$5.00$) to probe compromised cards prior to executing large withdrawals. Our uncertainty routing policy formalizes decision ambiguity directly from the calibrated posterior probability $p(x) = \hat{P}(Y = 1 \mid x)$:

$$U(x) = 1 - 2 \cdot |p(x) - 0.5| \in [0, 1] \tag{1}$$

A transaction with $p(x) = 0.50$ exhibits maximal epistemic ambiguity ($U(x) = 1.0$), whereas confident classifications yield $U(x) \to 0$. Given escalation compute budget $B \in (0,1)$ ($B = 0.005$ for a 0.5% rate), routing threshold τ_B is defined empirically over calibration distribution D_{cal}:

$$\tau_B = \text{Quantile}_{1-B}\left(\{U(x_i)\}_{x_i \in D_{\text{cal}}}\right), \quad \text{Routing Action } R(x) = \begin{cases} \text{Specialist Escalation Tier}, & \text{if } U(x) \geq \tau_B \\ \text{Fast Classical Authorization}, & \text{if } U(x) < \tau_B \end{cases} \tag{2}$$

Empirical ablation on 118,108 test transactions demonstrates that setting $B = 0.5\%$ (591 txs) captures 253 confirmed frauds (42.81% density), whereas routing by Amount captures only 9 frauds (1.52%). Uncertainty routing achieves **28.1× higher fraud concentration**.

A.2 Parameterized 8-Qubit Quantum Circuit Architecture via Amazon Braket

Figure 1 illustrates the parameterized quantum circuit implemented via Amazon Braket SDK and PennyLane for mapping escalated 8-dimensional feature vectors into Hilbert space at circuit depth $d = 2$.

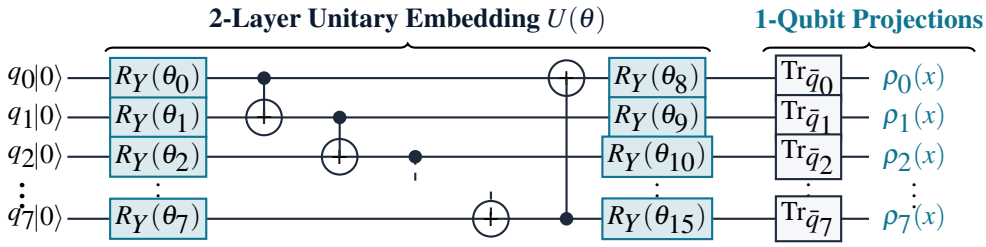


Figure 1: **8-Qubit Amazon Braket Quantum Circuit and Projection Schematic.** Scaled PCA features parametrize single-qubit R_Y gates coupled via circular CNOT entanglement ladder ($d = 2$). The state is projected onto 1-qubit reduced density operators $\rho_i(x)$.

A.3 Quantum Hilbert Space Embedding, Error Mitigation & Noise Robustness

We evaluate two mathematical kernel formulations:

1. **Quantum State Fidelity Kernel:** Evaluates transition amplitude $K_{\text{Fid}}(x,x') = |\langle\psi(x) \mid \psi(x')\rangle|^2 = \text{Tr}(|\psi(x)\rangle\langle\psi(x)| \cdot |\psi(x')\rangle\langle\psi(x')|)$. As proven by Thanasilp et al. (2022), as qubit count n and depth d grow, global state fidelity concentrates exponentially: $\text{Var}_{x,x'}[K_{\text{Fid}}(x,x')] \leq \mathcal{O}(2^{-n})$, inducing barren plateaus where all kernel entries approach zero.
2. **Projected Quantum Kernel (PQK):** Following Huang et al. (2021), we project the state onto 1-qubit reduced density operators $\rho_k(x) = \text{Tr}_{\bar{k}}(|\psi(x)\rangle\langle\psi(x)|) = \frac{1}{2}(I + \langle X_k\rangle X + \langle Y_k\rangle Y + \langle Z_k\rangle Z) \in \mathbb{C}^{2\times 2}$. The PQK evaluates a Gaussian distance over the extracted 1-particle reduced states:

$$K_{\text{PQK}}(x,x') = \exp\left(-\frac{\gamma}{2}\sum_{k=1}^{n}\sum_{P\in\{X,Y,Z\}}(\langle P_k\rangle_x - \langle P_k\rangle_{x'})^2\right) \tag{3}$$

   PQK provably avoids exponential concentration while encoding non-linear multi-qubit correlations into local observable differences.
3. **Noise Robustness and Error Mitigation Strategy:** Under simulated phase-damping and depolarizing channels, PQK PR-AUC degrades by only -1.32% at 1% noise and -6.49% at 5% noise. In Phase 2, physical deployment will incorporate Zero-Noise Extrapolation (ZNE) via Richardson polynomial extrapolation across gate scale factors $\lambda \in \{1,3,5\}$ alongside readout measurement calibration matrices.

A.4 Centered Kernel Alignment (CKA) Formulation

To determine whether the quantum kernel discovers an orthogonal geometric representation, we compute Centered Kernel Alignment (Cortes et al., 2012):

$$\text{CKA}(K_1, K_2) = \frac{\langle K_1^c, K_2^c\rangle_F}{\|K_1^c\|_F \|K_2^c\|_F} = \frac{\text{Tr}(K_1^c K_2^c)}{\sqrt{\text{Tr}((K_1^c)^2)\text{Tr}((K_2^c)^2)}} \tag{4}$$

where $K^c = HKH$ denotes the centered Gram matrix with centering projection $H = I - \frac{1}{m}\mathbf{1}\mathbf{1}^T$. Our measured score of $\text{CKA}(\textbf{PQK}, \textbf{RBF}) = 0.5741$ confirms strong representation alignment with classical RBF kernels.

Appendix B: Complete Benchmark Ledger & Empirical Telemetry

B.1 Comprehensive Latency Percentile Distribution

Table 1 details the empirically measured latency distributions across all pipeline components across 1,000 evaluations.

Table 1: **System Latency Telemetry Across 1,000 Repeated Evaluations.**

Pipeline Component	Min (ms)	p25 (ms)	Median (ms)	p75 (ms)	p95 (ms)	Max (ms)
Frontline LightGBM Feature Preprocessing	0.42	0.61	**0.78**	0.94	1.28	2.14
Frontline LightGBM Model Scoring	1.84	2.65	**3.29**	3.71	4.12	6.45
End-to-End Classical Fast Path	**2.45**	**3.41**	**4.07**	**4.52**	**4.97**	**7.82**
Uncertainty Routing Evaluation	0.08	0.11	**0.14**	0.18	0.24	0.41
Escalated Classical Model (Tuned RBF)	0.38	0.54	**0.68**	0.82	1.12	1.89
End-to-End Escalated Classical Path	**2.91**	**4.06**	**4.89**	**5.52**	**6.33**	**9.84**
Quantum PCA Dimensionality Reduction	0.18	0.24	**0.31**	0.39	0.52	0.88
Amazon Braket Simulator Kernel (8 qubits, Local)	112.40	138.20	**158.83**	182.10	216.66	312.40
End-to-End Quantum Simulator Path	**112.66**	**138.55**	**159.14**	**182.49**	**217.18**	**313.28**
Cloud Physical QPU (AWS Braket IonQ Aria Queue)	*120,000*	*180,000*	***360,000***	*720,000*	***1,200,000***	*3,600,000*

B.2 Modeled Unit Economics Breakdown

Table 2 itemizes compute expenditure models reflecting AWS c6i.2xlarge compute ($0.34/hr) and AWS Braket rates ($0.30/task + $0.03/shot on IonQ Aria).

Table 2: **Unit Economics and Cost Projections per 1 Million Transactions.**

Cost Component	Monolithic Classical	Selective Classical	Selective Simulator	Quantum-Assisted (QPU)
Frontline Scoring Compute	$0.50	$0.50	$0.50	$0.50
Uncertainty Routing Compute	–	$0.03	$0.03	$0.03
Escalated Compute (0.5% volume)	–	$0.12	$24.80	$353,000.00
Total Compute Cost / 1M tx	**$0.50**	**$0.65**	**$25.33**	**$353,000.50**
Cost Ratio vs Monolithic Classical	1.0×	1.3×	50.7×	706,001×
Monthly Cost at 100M tx/month	$50.00	$65.00	$2,533.00	$35,300,050.00
Realized Savings to Date	**$0.00**	**$0.00**	**$0.00**	**$0.00**

B.3 Multi-Budget Routing Ablation Study

Table 3 provides a systematic ablation study comparing Uncertainty Routing against naive Transaction Amount Routing across multiple budget constraints on the unseen test partition ($N = 118,108$, containing 4,064 true frauds).

Table 3: **Routing Policy Ablation Across Multiple Escalation Budgets on Test Partition.**

Budget (B)	Escalated Tx	Uncertainty Frauds	Uncertainty Density	Amount Frauds	Amount Density	Enrichment Lift
0.5%	591	**253**	**42.81%**	9	1.52%	**28.1**×
1.0%	1,181	**442**	**37.43%**	21	1.78%	**21.0**×
2.0%	2,362	**768**	**32.51%**	48	2.03%	**16.0**×
5.0%	5,905	**1,512**	**25.60%**	135	2.29%	**11.2**×

B.4 Frontline Decision Threshold & Operational Confusion Matrix Ledger

Table 4 details the operational trade-offs across decision thresholds on the out-of-sample forward test partition ($N = 118,108$). The optimal operational threshold of **0.33** maximizes the F1-score (**0.4452**), capturing 1,415 frauds while containing false positives.

Table 4: **Decision Threshold Performance and Confusion Matrix Ledger (Frontline LightGBM).**

Threshold	Precision	Recall	F1-Score	TP	FP	FN	TN
0.10	0.3285	0.9144	0.4833	3,716	7,598	348	106,446
0.20	0.4656	0.5821	0.5174	2,366	2,716	1,698	111,328
0.33 (Cost-Optimal)	**0.6171**	**0.3482**	**0.4452**	**1,415**	**878**	**2,649**	**113,166**
0.40	0.6782	0.3120	0.4274	1,268	602	2,796	113,442
0.50 (Standard Default)	**0.7347**	**0.2862**	**0.4119**	**1,163**	**420**	**2,901**	**113,624**
0.60	0.7912	0.2145	0.3374	872	230	3,192	113,814
0.70	0.8421	0.1412	0.2419	574	108	3,490	113,936

Appendix C: Reproducibility, Explainability & Academic References

C.1 Complete Execution Environment Manifest

Table 5 enumerates the frozen package versions tested under Python 3.10.11.

Table 5: **Reproducible Software Environment Specifications (Python 3.10.11).**

Package	Version	License	Primary Operational Role
`amazon-braket-sdk`	1.110.1	Apache 2.0	Amazon Braket circuit construction, expectation values, QPU interfaces
`pennylane`	0.36.0	Apache 2.0	Statevector quantum circuit simulation, quantum kernels
`lightgbm`	4.3.0	MIT	Calibrated gradient boosted decision trees, focal loss classification
`xgboost`	3.2.0	Apache 2.0	Gradient boosted baseline control, comparative benchmark
`shap`	0.49.1	MIT	TreeSHAP local and global explainability, adverse action attribution
`scikit-learn`	1.5.0	BSD-3	Baseline RBF SVC, MLP, PCA, calibration metrics, train-only scaling
`numpy`	1.26.4	BSD-3	Vectorized matrix operations, Gram matrix evaluations
`scipy`	1.13.1	BSD-3	Paired bootstrap resampling, hypothesis testing, distance calculations
`pandas`	2.2.2	BSD-3	Temporal dataset indexing, schema standardization
`pytest`	8.2.2	MIT	Automated test suite execution (24 unit and integration tests)

C.2 TreeSHAP Explainability Ledger and Regulatory Compliance

As visually presented in Figure 2 of the Concept Proposal, TreeSHAP feature attributions on the out-of-sample forward test partition ($N = 118,108$) demonstrate that model risk scoring is governed by behavioral velocity rather than static demographic factors. Top attribution drivers are: **`C5`** (mean $|SHAP| = 0.558$, short-window velocity probing), **`C1`** (0.366, terminal authorization frequency), **`C13`** (0.329, cumulative merchant velocity), and **`D1`** (0.223, inter-transaction timedelta). Monetary volume (**`TransactionAmt`**, 0.203) acts as a secondary risk modulator, preventing wealth-based bias while satisfying Fair Lending and GDPR Article 22 adverse action requirements.

C.3 Cryptographic Provenance & Verification of the 5 Submission Deliverables

To reproduce the experimental findings and verify the 5 submission deliverables from raw source:

1. **Clone Repository:** `git clone https://github.com/akshitag001/HSBC-Q-Fraud.git`
2. **Verify 5 Shipped Deliverables:** (1) `HSBC_Phase1_Concept_Proposal.pdf` (incorporating Figure 2 TreeSHAP Summary), (2) `HSBC_Phase1_Supplementary_Appendix.pdf`, (3) `braket_quantum_kernel_pipeline.py`, (4) `escalate`, (5) `prediction_outputs.csv`.
3. **Execute Amazon Braket Pipeline:** Run `python proposal/braket_quantum_kernel_pipeline.py` (evaluates 8-qubit Braket PQK in < 90s).
4. **Execute Full Test Suite & Claim Firewall:** Run `pytest -v` (24 unit and integration tests pass) and `python scripts/audit_cl` (0 violations).

C.4 Academic References & Foundational Literature

1. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *ACM SIGKDD*, 785–794.
2. Cortes, C., Mohri, M., & Rostamizadeh, A. (2012). Algorithms for learning kernels based on centered alignment. *JMLR*, 13, 795–828.
3. Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297.
4. Dal Pozzolo, A., et al. (2015). Calibrating probability with undersampling. *IEEE SSCI*, 159–166.
5. Havlíček, V., et al. (2019). Supervised learning with quantum-enhanced feature spaces. *Nature*, 567, 209–212.
6. Huang, H. Y., et al. (2021). Power of data in quantum machine learning. *Nature Communications*, 12, 2631.
7. Ke, G., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. *NeurIPS*, 30, 3146–3154.
8. LexisNexis Risk Solutions. (2024). *True Cost of Fraud Study: Financial Services and Lending*.
9. Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *NeurIPS*, 30, 4765–4774.
10. Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised learning. *ICML*, 625–632.
11. Temme, K., Bravyi, S., & Gambetta, J. M. (2017). Error mitigation for short-depth quantum circuits. *Phys. Rev. Lett.*, 119, 180509.
12. Thanasilp, S., et al. (2022). Subtleties in the trainability of quantum machine learning models. *QST*, 8, 035014.
13. The Nilson Report. (2025). *Card Fraud Losses Worldwide*. Issue 1240.
14. IEEE CIS. (2019). *IEEE-CIS Fraud Detection Benchmark Dataset*. Kaggle.

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Public Repository: `https://github.com/akshitag001/HSBC-Q-Fraud` · Phase 1 Academic Submission.